QRM2

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Week 4

1 Movie budgets and profits

There is another argument going on in the movie studio. Bob claims that production budgets are getting out of hand, and that the studio should focus on making cheaper movies. Chantal disagrees. She tells Bob that "Every dollar we spend on movie production is more than offset by the increase in movie profits".

Step a

Set up a regression model to test Chantal's claim, and estimate it. That is, estimate:

$$Profits_i = \beta_0 + \beta_1 Budget_i + \varepsilon_i.$$

Print a summary of your estimated model. [2 points]

```
# Load the dataset and calculate profit in dollars
movies <- read.delim("movies5.tsv")
movies$profit <- movies$revenue - movies$budget

m1 <- lm(profit ~ budget, data = movies)
summary(m1)
##
## Call:
## lm(formula = profit ~ budget, data = movies)
##
## Residuals:
##        Min         1Q     Median         3Q        Max
## -387813422  -39322110     610960    8901755  950609114
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -3.307e+06  4.636e+06  -0.713    0.476
## budget       1.830e+00  8.798e-02  20.802   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 109900000 on 907 degrees of freedom
## Multiple R-squared:  0.323,  Adjusted R-squared:  0.3223
## F-statistic: 432.7 on 1 and 907 DF,  p-value: < 2.2e-16
```

Your Answer:

We estimated a linear regression with profit as the dependent variable and budget as the independent variable. Both are measured in dollars. The estimated model is: predicted profit = -3,307,162.73 + 1.8301 times budget. The model uses all 909 movies.

Step b

What is the estimated value of β_1 and how do you interpret it? [2 points]

```
coef(m1)["budget"]
##   budget
## 1.830122
```

Your Answer:

The estimated coefficient is 1.8301. A one-dollar higher budget is associated with approximately 1.83 dollars higher predicted profit, on average. Profit already equals revenue minus budget. This relationship does not prove that spending an extra dollar causes profit to increase by this amount.

Step c

Test for the null hypothesis that $\beta_1 \geq 0$. Report the p-value and state your conclusion. [2 points]

```
# Use the lower tail because the alternative is beta_1 < 0
t_value <- summary(m1)$coefficients["budget", "t value"]
p_value <- pt(t_value, df = df.residual(m1))
p_value
## [1] 1
```

Your Answer:

The null hypothesis is H0: beta_1 >= 0, and the alternative is H1: beta_1 < 0. We use a 5% significance level. The t-value is approximately 20.80, and the one-sided p-value is approximately 1. We therefore do not reject H0. There is no evidence of a negative relationship. This test does not establish that beta_1 is greater than 1.

Step d

Next, estimate the model:

$$\log(Profits_i) = \beta_0 + \beta_1 \log(Budget_i) + \varepsilon_i.$$

When creating the variables Log Profits and Log Budget, make sure that movies with a Revenue or Budget of zero are assigned the value "NA". Print a summary of your estimated model. [2 points]

```
# Leave zero revenue or budget as NA in both log variables
movies$log_profit <- NA
movies$log_budget <- NA
valid_budget <- movies$budget > 0 & movies$revenue > 0
movies$log_budget[valid_budget] <- log(movies$budget[valid_budget])

# Log profit also requires a positive profit
valid_profit <- valid_budget & movies$profit > 0
movies$log_profit[valid_profit] <- log(movies$profit[valid_profit])

m2 <- lm(log_profit ~ log_budget, data = movies)
summary(m2)
##
## Call:
## lm(formula = log_profit ~ log_budget, data = movies)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -7.2956 -0.6776  0.1751  0.8725  3.1581
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  7.64769    0.76937    9.94   <2e-16 ***
## log_budget   0.58680    0.04482   13.09   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.346 on 461 degrees of freedom
##   (446 observations deleted due to missingness)
## Multiple R-squared:  0.271,  Adjusted R-squared:  0.2695
## F-statistic: 171.4 on 1 and 461 DF,  p-value: < 2.2e-16
```

Your Answer:

We used natural logarithms. Both log variables are NA when revenue or budget is zero. Log profit is also NA when profit is zero or negative, because these values do not have a finite real logarithm.

The estimated intercept is 7.64769 and the coefficient of log budget is 0.58680. The model uses 463 movies with positive profit, revenue and budget. Its R-squared is 0.271, so it explains approximately 27.1% of the variation in log profit among these movies.

Step e

What is the estimated value of β_1 and how do you interpret it? [2 points]

```
coef(m2)["log_budget"]
## log_budget
##  0.5868046
```

Your Answer:

The estimated coefficient of log budget is 0.58680. Since this is a log-log model, a 1% higher budget is associated with approximately 0.59% higher profit. This relationship applies to the movies included in the model and does not prove that increasing a movie's budget causes its profit to increase.

Step f

Which model has better fit? The level-level model or the log-log model? Explain. [2 points]

```
c(level_level = summary(m1)$r.squared,
  log_log = summary(m2)$r.squared)
## level_level     log_log
##   0.3230012   0.2710488
```

Your Answer:

The first model explains about 32.3% of the variation in profit. The second model explains about 27.1% of the variation in log profit.

The first model has a higher R-squared, but this does not necessarily mean it fits better. One model uses profit, while the other uses log profit. The first model also uses 909 movies, compared with 463 in the second model. Because of these differences, we cannot decide which model fits better just by comparing their R-squared values.

Step g

Who do you think is correct? Bob or Chantal? What would you advise the movie studio to do? [2 points]

```
# Check whether the 95% confidence interval is above 1
confint(m1, "budget", level = 0.95)
##          2.5 %   97.5 %
## budget 1.65746 2.002784
```

Your Answer:

The level-level model supports Chantal's claim as a relationship: one dollar higher budget is associated with about 1.83 dollars higher profit. The usual 95% confidence interval for the coefficient is 1.66 to 2.00, entirely above 1. The log model also shows a positive relationship, but its coefficient measures percentage changes rather than dollars.

However, neither model proves that increasing a movie's budget causes higher profit. Genre, audience demand and other factors may affect both. We advise the studio to consider each movie's expected costs and revenue, rather than always choosing cheaper movies or assuming that every extra dollar will pay off.

2 First actor gender and movie budgets

Step a

Make a plot with a 95% confidence interval with the mean log of budget on the y-axis, and whether the first actor of the movie is male or female on the x-axis. What do you conclude? [2 points]

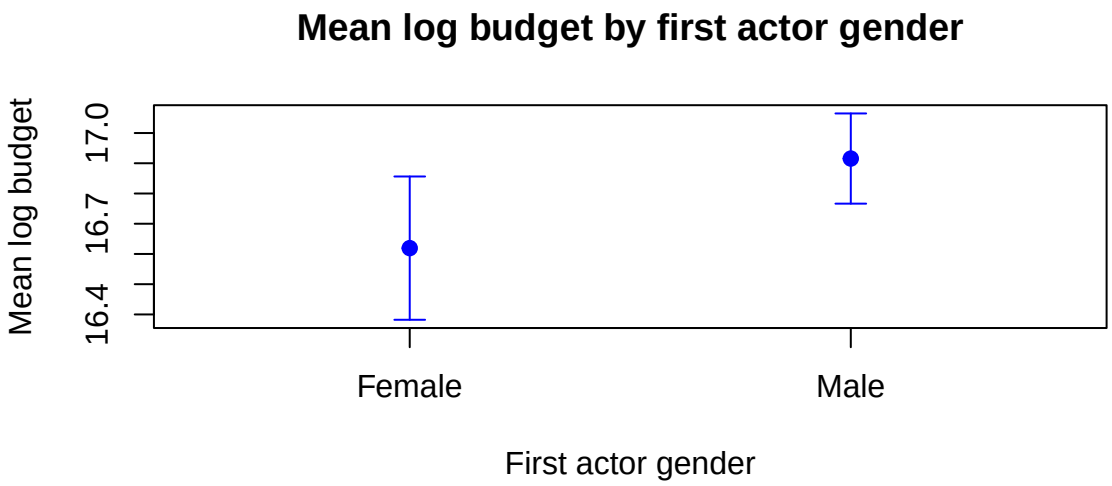
```
# For this question, use all positive budgets
movies$log_budget <- NA
positive_budget <- movies$budget > 0
movies$log_budget[positive_budget] <- log(movies$budget[positive_budget])

groups <- c("female", "male")
mean_log_budget <- numeric(2)
lower_budget <- numeric(2)
upper_budget <- numeric(2)

for (i in 1:2) {
  group_budget <- movies$log_budget[
    movies$first_actor_gender == groups[i]]
  group_budget <- group_budget[!is.na(group_budget)]
  mean_log_budget[i] <- mean(group_budget)
  interval <- t.test(group_budget, conf.level = 0.95)$conf.int
  lower_budget[i] <- interval[1]
  upper_budget[i] <- interval[2]
}

plot(1:2, mean_log_budget,
     main = "Mean log budget by first actor gender",
     xlab = "First actor gender", ylab = "Mean log budget",
     xaxt = "n", xlim = c(0.5, 2.5),
     ylim = range(lower_budget, upper_budget),
     pch = 19, col = "blue")
axis(1, at = 1:2, labels = c("Female", "Male"))

# Add the 95% confidence intervals
arrows(1:2, lower_budget, 1:2, upper_budget,
       angle = 90, code = 3, length = 0.08, col = "blue")
```



Your Answer:

The mean log budget is 16.62 for the female group and 16.92 for the male group. The 95% confidence intervals are approximately 16.38 to 16.86 and 16.77 to 17.06. The male group has a higher mean log budget. The intervals overlap, but this alone does not tell us whether the difference is statistically significant.

For this separate question, we use all movies with a positive budget and a recorded first actor gender. We do not require positive revenue, because this model only concerns budget and gender.

Step b

Estimate the following simple OLS model:

$$\log(budget_i) = \beta_0 + \beta_1(FirstActorMale)_i + \varepsilon_i.$$

Is the estimated coefficient for β_1 significantly different from zero? How do you interpret its estimate, and how does this relate to your conclusion in 2a? [2 points]

```
# Code male as 1 and female as 0; missing values remain NA
movies$first_actor_male <- as.numeric(movies$first_actor_gender == "male")
m3 <- lm(log_budget ~ first_actor_male, data = movies)
summary(m3)
##
## Call:
## lm(formula = log_budget ~ first_actor_male, data = movies)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -16.9156  -0.5351   0.2872   0.9943   2.4945
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)       16.6193     0.1280 129.884   <2e-16 ***
## first_actor_male   0.2963     0.1480   2.002   0.0457 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.707 on 702 degrees of freedom
##   (205 observations deleted due to missingness)
## Multiple R-squared:  0.005674,   Adjusted R-squared:  0.004258
## F-statistic: 4.006 on 1 and 702 DF,  p-value: 0.04572

# Convert the log difference into a percentage difference
100 * (exp(coef(m3)["first_actor_male"]) - 1)
## first_actor_male
##          34.4858
```

Your Answer:

The coefficient is 0.2963, with a p-value of approximately 0.0457. We reject H0: beta_1 = 0 against H1: beta_1 is not zero at the 5% significance level. The model uses 704 movies: 178 in the female group and 526 in the male group.

The male group has a mean log budget that is 0.2963 higher. This corresponds to a geometric mean budget about 34.49% higher. The direction agrees with the plot in step a. The regression finds a significant difference even though the separate confidence intervals overlap. It does not prove that the first actor's gender causes higher budgets.

Step c

Now, have a close look at your data frame. Can you find any instances of male first actors who are wrongly labelled as being female, or vice versa? What would such mislabelling mean for the coefficient you estimated under 2b? [2 points]

```
# Show two examples of incorrect labels in the original data
unique(movies[movies$first_actor %in% c("Morgan Freeman", "Jackie Chan"),
              c("first_actor", "first_actor_gender")])
##        first_actor first_actor_gender
## 33  Morgan Freeman             female
## 453    Jackie Chan             female
```

Your Answer:

Morgan Freeman and Jackie Chan are male actors but are labelled female in the dataset. Incorrect labels put movies in the wrong group and can bias the coefficient. If the mistakes are unrelated to budget within each true gender group, they generally make the estimated difference smaller. Otherwise, the bias can go either way. The results above use the original labels, so we should check them before drawing strong conclusions.

Week 5

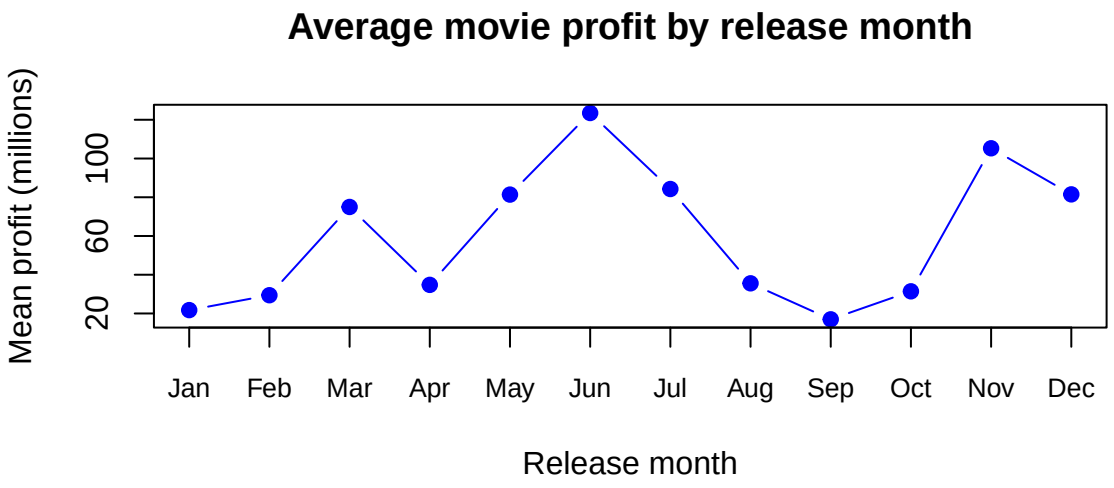
Movie release month and revenue predictions

Step a

Create a plot of the mean profits by month of release. Do you see any indication that month of release matters to the profits of the movie? [2 points]

```
# Calculate the mean profit for each release month
profits_by_month <- aggregate(profit ~ release_month,
                              data = movies, FUN = mean)

plot(profits_by_month$release_month,
     profits_by_month$profit / 1000000,
     type = "b", pch = 19, col = "blue", xaxt = "n",
     main = "Average movie profit by release month",
     xlab = "Release month", ylab = "Mean profit (millions)")
axis(1, at = 1:12, labels = month.abb, cex.axis = 0.8)
```




```
profits_by_month[which.max(profits_by_month$profit), ]
##   release_month    profit
## 6             6 123486518
```

Your Answer:

Mean profit differs across months. June has the highest mean (~$123.49M) and September the lowest (~$16.98M). The fluctuation suggests release month matters, but the plot alone doesn't prove causality, as movies released in different months may also differ in budget and genre.

Step b

Estimate an OLS model which has as dependent variable the log of profits of a movie, and as independent variables the log of budget, a dummy for whether the movie was released in English or not, and a linear term for the month of release. Show a summary of the resulting model and interpret each coefficient. [4 points]

```
# Code English as 1 and other languages as 0
movies$english <- as.numeric(movies$original_language == "en")

# Use the same complete sample for the models in steps b to e
model_data <- movies[complete.cases(movies[,
  c("log_profit", "log_budget", "english", "release_month")]), ]

m5b <- lm(log_profit ~ log_budget + english + release_month,
          data = model_data)
summary(m5b)
##
## Call:
## lm(formula = log_profit ~ log_budget + english + release_month,
##     data = model_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -7.2036 -0.6992  0.1714  0.8886  2.9585
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    6.96828    0.81336   8.567   <2e-16 ***
## log_budget     0.57697    0.04488  12.857   <2e-16 ***
## english        0.72135    0.34411   2.096   0.0366 *
## release_month  0.02106    0.01836   1.147   0.2519
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.34 on 459 degrees of freedom
## Multiple R-squared:  0.2807, Adjusted R-squared:  0.276
## F-statistic: 59.72 on 3 and 459 DF,  p-value: < 2.2e-16
```

Your Answer:

The model uses 463 movies. Holding other variables constant: Intercept (6.9683): Predicted log profit for a non-English movie with a $1 budget and release month zero (practically meaningless). Log Budget (0.5770): A 1% higher budget is associated with about 0.58% higher profit. English (0.7214): English movies have a predicted log profit 0.7214 higher (~105.72% higher geometric mean profit). Release Month (0.0211): Moving one month later is associated with about 2.13% higher profit. These are associations, not causal effects.

Step c

Test for the hypothesis that the coefficient that belongs to month of release is zero. [2 points]

```
summary(m5b)$coefficients["release_month", ]
##   Estimate Std. Error    t value   Pr(>|t|)
## 0.02106297 0.01836138 1.14713482 0.25192350
```

Your Answer:

The null hypothesis is H0: beta_month = 0. With t = 1.147 and p = 0.2519, we do not reject H0 at the 5% level. There is no statistically significant linear month trend in this model, though a non-linear pattern is not ruled out.

Step d

Based on your plot in a, do you consider the choice that month of release enters the model linearly under b reasonable? Estimate a specification that allows for a more flexible curve. In this new specification, test for the null hypothesis that month of release does not impact profits. This might require testing multiple terms at once. [4 points]

```
# Treat month as a category, with January as the reference month
m5d <- lm(log_profit ~ log_budget + english + factor(release_month),
          data = model_data)
summary(m5d)$coefficients
##                              Estimate Std. Error    t value     Pr(>|t|)
## (Intercept)              7.65331054 0.84503788  9.0567663 4.140274e-18
## log_budget               0.53453882 0.04786166 11.1684143 1.005551e-25
## english                  0.79437367 0.34582313  2.2970518 2.207515e-02
## factor(release_month)2  -0.41090814 0.38084883 -1.0789271 2.811995e-01
## factor(release_month)3  -0.04083193 0.32203765 -0.1267924 8.991615e-01
## factor(release_month)4  -0.27005408 0.34353186 -0.7861107 4.322173e-01
## factor(release_month)5   0.49839240 0.32407370  1.5378983 1.247779e-01
## factor(release_month)6   0.59488897 0.31297945  1.9007285 5.797770e-02
## factor(release_month)7   0.15807347 0.31904346  0.4954606 6.205174e-01
## factor(release_month)8   0.15656912 0.30236677  0.5178119 6.048445e-01
## factor(release_month)9  -0.21889686 0.29299624 -0.7470978 4.553955e-01
## factor(release_month)10  0.09947628 0.29975639  0.3318571 7.401520e-01
## factor(release_month)11  0.43306818 0.31456575  1.3767175 1.692857e-01
## factor(release_month)12  0.16913548 0.29218396  0.5788664 5.629696e-01

# Jointly test all 11 month coefficients using the same movies
m5d_restricted <- lm(log_profit ~ log_budget + english,
                     data = model_data)
anova(m5d_restricted, m5d)
## Analysis of Variance Table
##
## Model 1: log_profit ~ log_budget + english
## Model 2: log_profit ~ log_budget + english + factor(release_month)
##   Res.Df    RSS Df Sum of Sq      F  Pr(>F)
## 1    460 826.10
## 2    449 792.83 11    33.267 1.7127 0.06788 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Your Answer:

A linear trend poorly captures the seasonal rise and fall, so we treat month categorically with January as the reference category. To test whether months jointly impact profits (H 0 : all 11 month coefficients equal zero), the joint F-test gives F(11,449)=1.71 (p=0.0679). We do not reject H 0 at the 5% level; the evidence is insufficient to conclude that months jointly improve the model after controlling for budget and language.

Step e

One executive at the studio wants to time the release of the movie to a specific month of the year such that they can maximize revenue. Based on your model under d, what would you advise the movie studio regarding the timing of the release of the movie? [2 points]

```
# Compare months while keeping budget and language the same
month_options <- data.frame(log_budget = mean(model_data$log_budget),
                            english = 1, release_month = 1:12)
predicted_log_profit <- predict(m5d, newdata = month_options)
month.name[which.max(predicted_log_profit)]
## [1] "June"
```

Your Answer: June yields the highest predicted log profit holding budget and language constant. However, because the joint test in step d is not significant at 5%, we cannot confidently recommend June as the definitive optimal month. Furthermore, the model predicts log profit rather than revenue, and relies on historical data. Studios should weigh June against competition, audience demand, and genre.

Step f

The movie studio that you work at is releasing a new movie in 2026. It will be an English-spoken Thriller movie with a budget of 40,000,0000.

Estimate a model that is able to predict the revenue of this movie. Give its predicted revenue and include a 99% prediction interval. [6 points]

```
# Predict revenue, including movies that did not make a profit
revenue_data <- subset(movies, revenue > 0 & budget > 0)
revenue_data$log_revenue <- log(revenue_data$revenue)
revenue_data$thriller <- as.numeric(revenue_data$genre == "Thriller")

m5f <- lm(log_revenue ~ log_budget + english + thriller + release_year,
          data = revenue_data)
summary(m5f)
##
## Call:
## lm(formula = log_revenue ~ log_budget + english + thriller +
##     release_year, data = revenue_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.8618  -0.6587   0.1927   0.7990   6.5821
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -30.280042  18.025919  -1.680   0.0935 .
## log_budget     0.878029   0.043411  20.226   <2e-16 ***
## english        0.361804   0.317094   1.141   0.2543
## thriller       0.029167   0.159035   0.183   0.8545
## release_year   0.016263   0.008977   1.812   0.0705 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.477 on 624 degrees of freedom
## Multiple R-squared:  0.4062, Adjusted R-squared:  0.4024
## F-statistic: 106.7 on 4 and 624 DF,  p-value: < 2.2e-16

# Use a budget of 40 million, as in our original code
new_movie <- data.frame(log_budget = log(40000000), english = 1,
                        thriller = 1, release_year = 2026)

# Predict one movie and convert the result back to dollars
prediction_log <- predict(m5f, newdata = new_movie,
                          interval = "prediction", level = 0.99)
exp(prediction_log)
##         fit     lwr        upr
## 1 100779498 2111898 4809184565
```

Your Answer:

Using a $40M budget, English language, Thriller genre, and 2026 release year across 629 movies: Predicted Revenue: Approximately $100.78 million (median prediction). 99% Prediction Interval: Approximately $2.11 million to $4,809.18 million. The interval is extremely wide, reflecting high uncertainty. This relies on log-normal error assumptions and assumes relationships from 1990–2016 hold in 2026.